Institute of Systems Neuroscience,
University Medical Center
Hamburg-Eppendorf, Hamburg, Germany

LOVE REPLICATIONS WEEK

LOVE REPLICATIONS WEEK — MARCH 2-6

Maren Klingelhöfer-Jens

Getting started: Writing reproducible manuscripts using R Markdown

IGOR
Biological Psychology and Neuropsychology

RN DE German Reproducibility Network

Universitätsklinikum Hamburg-Eppendorf

DGPs
Fachgruppe Biologische Psychologie und Neuropsychologie

Interest Group for Open and Reproducible Science (IGOR)

Formation: Spring 2020
Members: 125 (from 05/2025)

Speakers since 2025: Gordon Feld & Maria Bruntsch

Activities

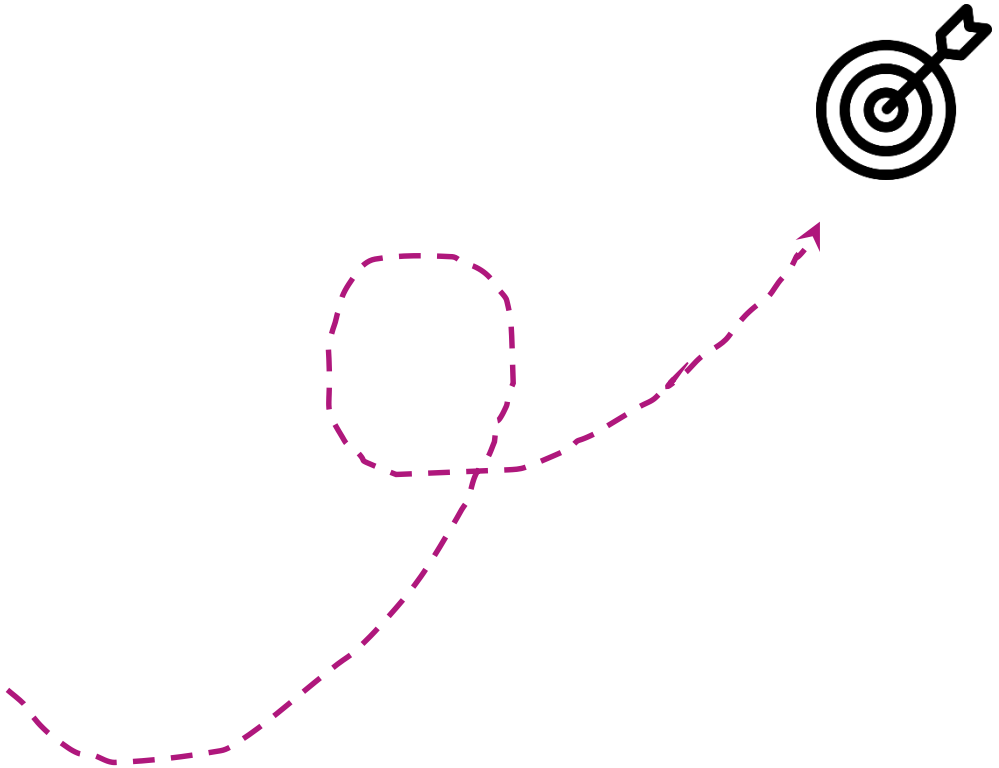
- Member Node of the ‚German Reproducibility Network‘ (GRN) since 2021; since 05/2022 seat in the GRN-Steering
- **Precision in Human Neuroscience** (*Elife*, Leads: Mario Reutter, Stephan Nebe, Gordon Feld, & Tina Lonsdorf)
- **Two Online-Hackathons and four active projects** about Data Transformation, Power analysis, EEG Analysis & Sustainability
- At **PuG conference**: roundtable, price, social evening
- **BIDS consultation** hours

Join/follow/connect with us!

@igor-dgps.bsky.social

OUTLINE

- The idea
- Should I?
- Prerequisites
- R Markdown demonstration
- Useful packages
- Where to share?
- Collaborative writing
- More reproducibility stuff
- Resources

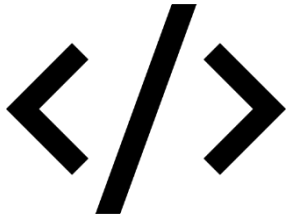


THE IDEA

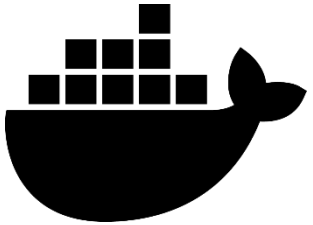
(Computational) Reproducibility:

Make

raw data



code and documentation



computational environment

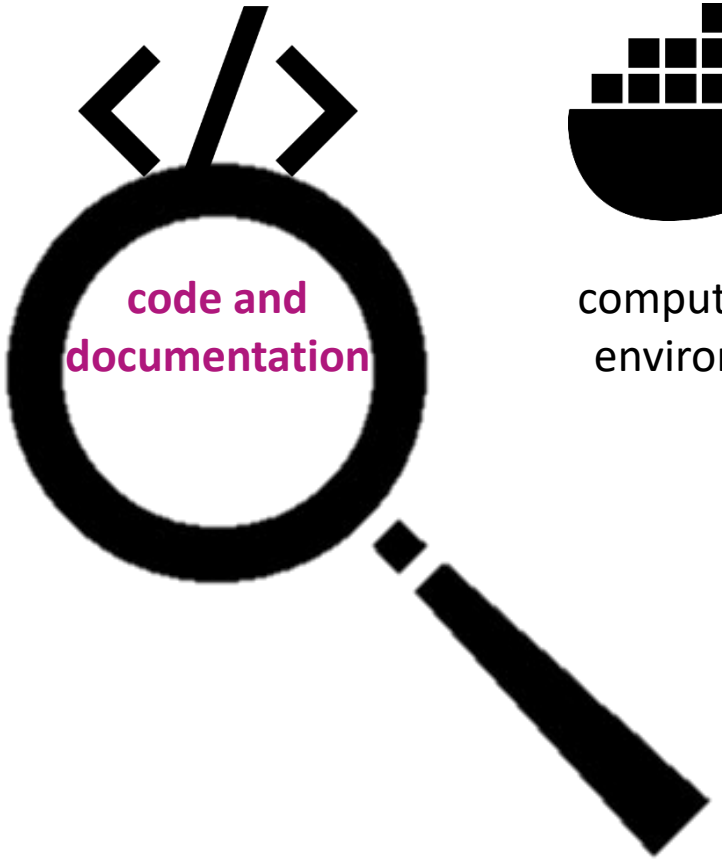
available and accessible

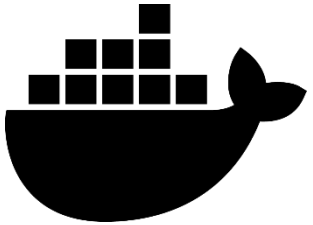
THE IDEA

(Computational) Reproducibility:

Make

raw data





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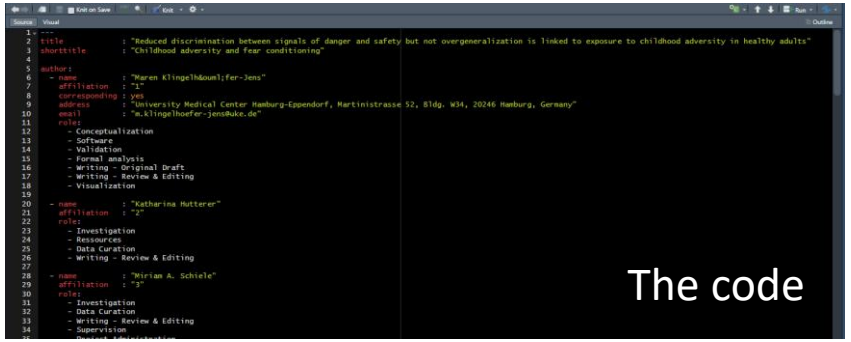
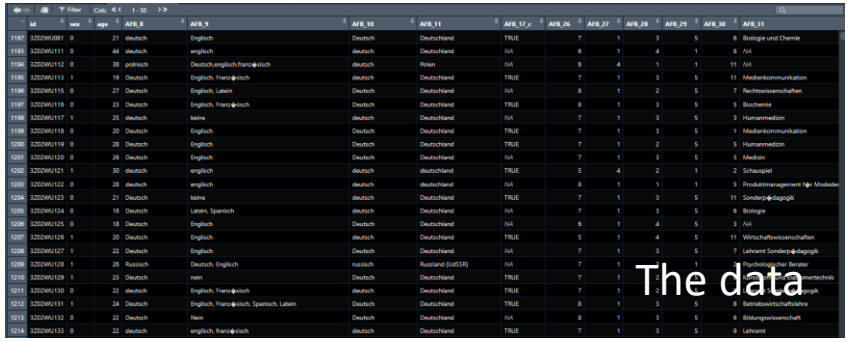
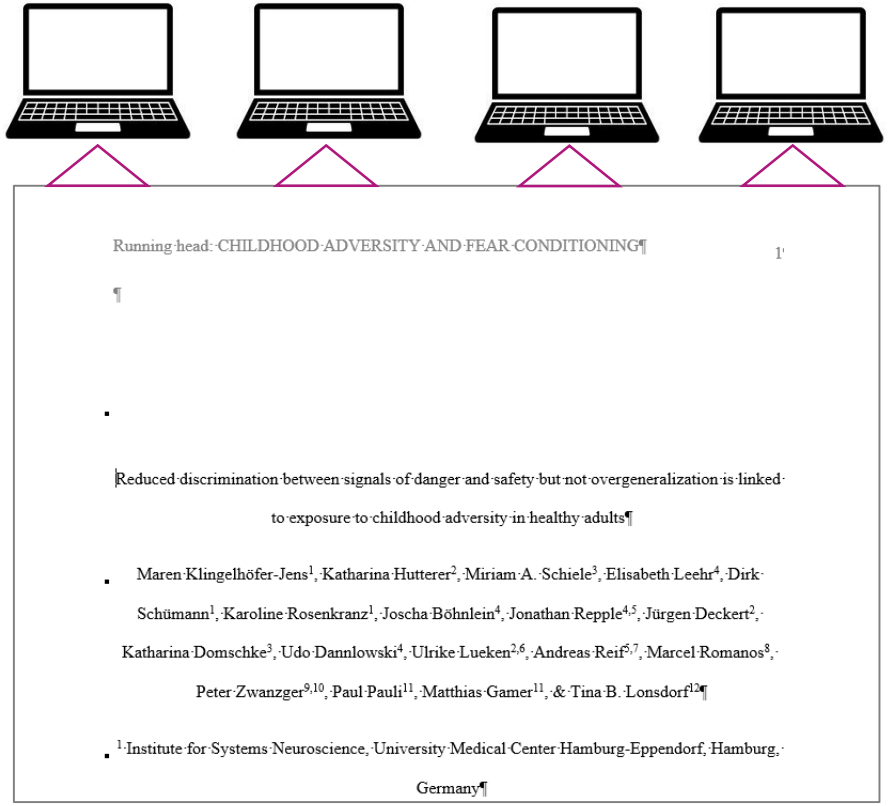
THE IDEA


---
title : "Reduced discrimination between signals of danger and safety but not overgeneralization is linked to exposure to childhood adversity in healthy adults"
shorttitle : "Childhood adversity and fear conditioning"
author:
- name : "Maren Klingelhöfer-Jens"
affiliation : "1"
corresponding : yes
address : "University Medical Center Hamburg-Eppendorf, Martinistrasse 52, Bldg. W34, 20246 Hamburg, Germany"
email : "m.klingelhoefer-jens@uke.de"
role:
- Conceptualization
- Software
- Validation
- Formal analysis
- Writing - Original Draft
- Writing - Review & Editing
- Visualization
- name : "Katharina Hutterer"
affiliation : "2"
role:
- Investigation
- Ressources
- Data Curation
- Writing - Review & Editing
- name : "Miriam A. Schiele"
affiliation : "3"
role:
- Investigation
- Data Curation
- Writing - Review & Editing
- Supervision
The code


The data


Running head: CHILDHOOD ADVERSITY AND FEAR CONDITIONING
1
Reduced discrimination between signals of danger and safety but not overgeneralization is linked to exposure to childhood adversity in healthy adults
Maren Klingelhöfer-Jens[1], Katharina Hutterer[2], Miriam A. Schiele[3], Elisabeth Leehr[4], Dirk Schümann[1], Karoline Rosenkranz[1], Joscha Böhnlein[4], Jonathan Repple[4,5], Jürgen Deckert[2], Katharina Domschke[3], Udo Dannlowski[4], Ulrike Lueken[2,6], Andreas Reif[5,7], Marcel Romanos[8], Peter Zwanzger[9,10], Paul Pauli[11], Matthias Gamer[11], & Tina B. Lonsdorf[12]
[1] Institute for Systems Neuroscience, University Medical Center Hamburg-Eppendorf, Hamburg, Germany

SHOULD I?

30% could not be reproduced!

The Reproducibility of Statistical Results in Psychological Research: An Investigation Using Unpublished Raw Da

Richard Artner, Thomas Verliefde, Sara Steegen, Sara Gomes, Frits Tra Francis Tuerlinckx, and Wolf Vanpaemel
Faculty of Psychology and Educational Sciences, KU L en

7% not significant anymore!

Time-consuming trial and error work!

Abstract
estigated the reproducibility of the major statistical conclusion drawn in 46 articles published in 2012 in APA journals. After having identified 232 key statistical claims, we tried to reproduce, for each claim, the test statistic, its degrees of freedom, and the corresponding *p* value, starting from the raw data that were provided by the authors and closely following the Method section in the article. Out of the 232 claims, we were able to successfully reproduce 163 (70%), 18 of which only by deviating from the article's analytical description. Thirteen (7%) of the 185 claims deemed significant by the authors are no longer so. The reproduction successes were often the result of cumbersome and time-consuming trial-and-error work, suggesting that APA style reporting in conjunction with raw data makes numerical

SHOULD I?

- Just one exampe: Artner et al., 2021
- Important prerequisite for robustness and replicability
- Helps avoid transcription and rounding errors
- Maximum transparency
- Protocol for your future self
- Imagine you have to exclude one participant – after finishing the manuscript

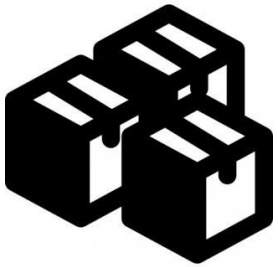
PREREQUISITES

Install:

- Latest R and R Markdown version
- R packages: *citr* (Aust, 2019), *papaja* (Aust, 2023)
- MiKTeX
- Zotero + Better Bib(La)TeX Plugin for Zotero

Let‘s switch to

USEFUL PACKAGES



Figures:

- *ggplot2* (Wickham, 2016)

Combine figures:

- *patchwork* (Pedersen, 2024)
- *cowplot* (Wilke, 2024)
- *ggpubr* (Kasssambra, 2023)

Tables:

- *kableExtra* (Zhu, 2024)
- *flextable* (Gohel & Skintzos, 2024)
- *papaja: apa_table()*

Other:

- *Rticles* (Allaire et al., 2024)

SHARE HERE

zenodo

OSF

GitHub

REALITY CHECK

- There will be pain
- It takes time and effort
- There will be problems, but many solutions
- It is worth it!
- It is also fun and rewarding!
- Reproducibility is not a dichotomy, it is a continuum

Nobody – not even yourself – can reproduce your analysis

Push a button that allows everyone to reproduce everything from raw data to published work

COLLABORATIVE WRITING

We simply use **Google Docs** and transfer the code to

R Markdown

R packages:

- *rmdrive*
- *redoc package*

Introduction

Self-report measures are commonly used both in the clinics and for research purposes to assess symptoms of depression and anxiety. The by far most cited questionnaire for measuring trait anxiety is the Spielberger State-Trait Anxiety Inventory (STAI) - a self-administered questionnaire, developed by Charles Spielberger [@spielberger1970; @spielberger1983], that consists of two scales with 20 items each.

The state (S) anxiety scale refers to the current state of anxious apprehension, tension, worry, nervousness, as well as autonomic nervous system arousal ("how do you feel right now?"). Criterion validity of this scale is indicated by healthy individuals scoring higher on the state scale during stressful situations (e.g., exams) than during non-stressful situations [@spielberger1983; @spielberger1994].

MORE REPRODUCIBILITY

For more reproducibility, use…

- Git and GitHub for version control (https://happygitwithr.com/)
- R package renv for reproducible environments (https://rstudio.github.io/renv/articles/renv.html)

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- Docker as a container (www.docker.com)

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- Docker as a container (www.docker.com)
- CodeOcean and (https://codeocean.com/)
- Binder as virtual capsules (https://mybinder.org/)

RESOURCES

Reproducible manuscript writing:
https://poldrack.github.io/psych-open-science-guide/2_codesharing.html
https://psyteachr.github.io/reprores-v3/
https://www.youtube.com/watch?v=JA-vLsN-sic
https://www.youtube.com/watch?v=-XHE21XzZso

Reproducible workflow:
https://qcmb.psychopen.eu/index.php/qcmb/article/download/3763/3763.html?inline=1

R Markdown:
https://bookdown.org/yihui/rmarkdown/
https://www.markdowntutorial.com/
https://resulumit.com/teaching/rmd_workshop.html#1

Coding:
The Art of Readable Code (Boswell & Foucher)

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Aust, F., & Barth, M. (2023). *papaja: Prepare reproducible APA journal articles with R Markdown*. Retrieved from https://github.com/crsh/papaja

Gohel, D., & Skintzos, P. (2024). *Flextable: Functions for tabular reporting*. Retrieved from https://CRAN.R-project.org/package=flextable

Kassambara, A. (2023). *Ggpubr: 'ggplot2' based publication ready plots*. Retrieved from https://CRAN.R-project.org/package=ggpubr

Pedersen, T. L. (2024). *Patchwork: The composer of plots*. Retrieved from https://CRAN.R-project.org/package=patchwork

Wickham, H. (2016). *ggplot2: Elegant graphics for data analysis*. Springer-Verlag New York. Retrieved from https://ggplot2.tidyverse.org

Wilke, C. O. (2024). *Cowplot: Streamlined plot theme and plot annotations for 'ggplot2'*. Retrieved from https://CRAN.R-project.org/package=cowplot

Zhu, H. (2024). *kableExtra: Construct complex table with 'kable' and pipe syntax*. Retrieved from https://CRAN.R-project.org/package=kableExtra

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Get in touch!

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THANK YOU!

☺

Lonsdorflab.com

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